

Open Source Lecture 4

1-Vectors in games are used to store ____ .

- a- Positions
- b- Directions
- c- Velocities
- d- All above

Answer : d. note (“ mostly physics “)

2-The is the magnitude of (4,3) .

- a- 7
- b- 25
- c- 5
- d- 0

Answer : c , note (“ it’s the vector lengths we use Pythagorean theorem “)

3- Find the distance between (1,2) and (3,3) .

- a- 2
- b- 2.25
- c- 2.23
- d- 2.24
- e- 0

Answer : c. note (we can get the distance by subtraction and then get the Magnitude)

4-A vector with length of two is called normalized.

- a- True
- b- False

Answer : b, explanation (its 1 and we divide each componenet by the vector length to get 1)

5-at the dot product if they’re perpendicular the dot product is zero.

- a- True
- b- False

Answer : a.

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6-Each iteration of the game loop is known as a pixel .

- a- True
- b- False

Answer : b, explanation (" its called a frame ")

7- The traditional game loop is broken into three distinct phases , processing inputs , updating the game world and generating outputs.

- a- True
- b- False

Answer : a.

8-For real time games the progression of time is typically measured in fractions of seconds

- a- True
- b- False

Answer : a.

9-Concept of ____ is the amount of elapsed game time since the last frame .

- a- Delta time
- b- Mansoura time
- c- Dakahliya time
- d- All above

Answer : a, note ("it was a joke sry ")

10- if the ideal movement speed is 150 pixels per second the pseudocode would be .

- a- `enemy.position.x += 150 + deltaTime`
- b- `enemy.position.x += 150/deltaTime`
- c- `enemy.position.x += 150*deltaTime`
- d- all above

answer : c.

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11- A sprite is a 2D visual object within the game world that can be drawn using a single image .

- a- True
- b- False

Answer : a. note (“ use sprites efficiently “)

12-in _____ algorithm all the sprites in a scene are sorted from back to front based on Z order/depth Values..

- a- Painter's
- b- Viewer's
- c- Both
- d- None

Answer : a.

13- We cant put every sprite in a file instead we use a _____ to put all sprites in a single file .

- a- Sprite storage
- b- Sprite sheet
- c- Sprite zero
- d- None

Answer : b.

14-Tile maps technique is for creating a game world out of modular building blocks

- a- True
- b- False

Answer : a, note (“ its useful for repeating patterns “)

15- Collision is almost used in every game for hits , which hit type is quite super precise collision detects.

- a- Pixel perfect
- b- Hitbox
- c- Multiple hit boxes
- d- None

Answer : a.

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16- Which type of these hitboxes have a better performance.

- a- Pixel perfect
- b- Hitbox
- c- Multiple hit boxes
- d- None

Answer : b , note (“ you know think about it like the more details the more power you need “)

17-Which bound box type rotates with object rotation .

- a- Axis -Aligned Bounding Box
- b- Oriented Bounding Box
- c- None
- d- Both

Answer : b.

18- ____ can be thought of as an AABB with two semicircles attached to the top and bottom.

- a- Capsule
- b- Box
- c- Rectangle
- d- None

Answer : a.

19-_____ refers to when a collection of 2D sprites are made to move dependently.

- a- Paralanaguage
- b- Parallaxing
- c- Paralizm
- d- None

Answer : b.

20- Isometric means non equal measures.

- a- True
- b- False

Answer : b, explanation (“ its Greek origin and means equal measures “)

“Thanks”